Common Investigations in Neonatology

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Common Investigations in Neonatology

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Outline

Common investigations · CBC & blood film · Blood gas · Transfusion thresholds (red cells, platelets) · White cell count & ANC · CRP & procalcitonin · Blood culture & lumbar puncture · Electrolytes (Na, K) · Bone profile (Ca, PO , Mg) · Thyroid profile · Bilirubin & liver function · Imaging

Common investigations in neonatology

Blood glucose · blood gas analysis · complete/full blood count · peripheral blood smear · blood culture · metabolic panel (renal function, electrolytes) · liver function tests · bone profile (calcium, phosphorus, alkaline phosphatase, vitamin D, PTH) · thyroid profile · urinalysis & culture · lumbar puncture.

Complete blood count (CBC)

Typically part of the admission work-up; further requests depend on the clinical condition and initial results.

- **Hb, Hct**
- **Platelet count**
- **RBC count** – a low count indicates the state of marrow erythropoiesis
- **Reticulocyte count** – supports a diagnosis of blood loss or haemolysis and helps decide on transfusion
- **MCV** – a **low MCV** is seen in iron-deficiency anaemia, thalassaemia and ABO incompatibility
- **MCHC, MCH**

Blood gas analysis

Evaluates **pH, paO and bicarbonate** to assess respiratory function and metabolic acidosis; often drawn **preductal** to measure ductal shunting.

Peripheral blood film

Important in any baby with **unexplained anaemia, thrombocytopenia or other concerns** (e.g. leukaemoid reaction in trisomy 21).

> Automated counters may **exaggerate the white cell count** – newborns have many **nucleated red cells** that can be miscounted as WBCs – so **manual verification** by film review matters.

Packed cell (red cell) transfusion – thresholds

Hb (g/dl) / Hct (%) thresholds for PRBC transfusion in preterm neonates $<$ **32 weeks**:

Postnatal age	Ventilated	O /NIPPV	No support
First 24 h	12.0 (35)	12.0 (35)	10.0 (30)
Week 1	12.0 (35)	10.0 (30)	10.0 (30)
Week 2	10.0 (30)	9.5 (28)	7.5 (23)
Week 3+	10.0 (30)	8.5 (25)	7.5 (23)

> The sicker the baby (more respiratory support), the **higher** the Hb threshold to transfuse.

Platelets

- Sampling technique matters – **clumping loses platelets** and gives falsely low counts; collect the CBC sample first when free-flowing and mix gently.
- **Normal range 150-450 x10 /L (150-450,000).**

Platelet transfusion thresholds:

Platelet count	Indication
$<$ 25,000	Neonate with **no bleeding** (incl. NAIT with no bleeding/no family ICH)

Platelet count	Indication
$<$ 50,000	Bleeding, coagulopathy, before surgery, or NAIT with previously affected sibling (ICH)
$<$ 100,000	Major bleeding or major surgery (e.g. neurosurgery)

White cell count & differential

- Highly variable and **not very specific**; general guide **5,000-30,000/mm³** (preterm 6,000-19,000; term 10,000-26,000).
- **Very low WBC is more concerning than a high WBC.**
- **Absolute neutrophil count (ANC): $<$ 1500 is low, $<$ 500 is very low** – both suggest risk of sepsis.
- **Band cells** = immature neutrophils with a C-shaped (non-segmented) nucleus.
- **Immature:total (I:T) neutrophil ratio: 0-0.2 normal**, **0.2-0.25 suggestive of infection**, higher = greater risk.

CRP (C-reactive protein)

- Acute-phase protein made in the **liver** in response to inflammation (infection, vaccination, aspiration, surgery, extravasation).
- **Normal $<$ 6.10 mg%**; a higher, sustained rise (**$>$ 40 mg%**) is seen in bacterial sepsis.
- **Lag of 12-24 hours** before it rises; some extreme preterm babies don't mount a response. The **trend** guides duration of treatment.

Serum procalcitonin

Another inflammatory marker – more expensive, with a **quicker rise, higher peak and quicker fall**. **Not reliable in the first 1-3 days** (wide, often higher range). Useful when clinical features fit sepsis but **CRP is normal**.

Blood culture

- Take **0.5-1 ml** with full aseptic precautions (sterile gloves, chlorhexidine skin prep).
- Most labs report by **36 hours**; if the baby is stable with normal markers and a negative culture, antibiotics can often stop by then.
- A positive culture – treatment duration depends on the organism and CSF findings; **rationalise antibiotics** on culture and sensitivity.

Lumbar puncture

- Indicated in **early-onset sepsis** in all babies with high CRP, positive blood culture or neurologic symptoms; in **late-onset sepsis** as part of the sepsis screen.
- **Only exception** to LP with a positive culture: **CoNS (coagulase-negative staph) sepsis.**
- Consider **herpes PCR** in any newborn with rash and sepsis features, or high fever with encephalopathy – **treat with aciclovir while awaiting results.**

Serum electrolytes

Sodium – normal **135-145 meq/L**.

- **Hyponatraemia** – significant if < 130, urgent below 125, dangerous below 120. **SIADH** is a common cause (treat with **fluid restriction**); also low intake and excess loss.
- **Hypernatraemia** – usually from **increased free-water loss** (insensible loss in extreme preterm) or **dehydration** from inadequate feeding in term babies. Correct the free-water deficit **very slowly** so sodium doesn't drop more than **0.5 meq/L per hour** (avoid brain injury).

Potassium – normal **3.5-5.5 meq/L** (higher in extreme preterm). Needed for cardiac/skeletal muscle contraction, acid-base balance and all cell functions; abnormal levels cause weakness, **arrhythmias/ECG changes, death.**

- **Hypokalaemia** – too little intake, renal losses (immaturity, polyuria, diuretics), GI losses.
- **Hyperkalaemia** – excess intake, haemolysis, bruising, acidosis, multiple transfusions.

Bone profile

Calcium – most abundant mineral; **99% in bone**. Term nadir **1.10-1.36 mmol/L at 24 hours**; ~30% develop **early hypocalcaemia** (first 2 days). Term hypocalcaemia = **Ca $<$ 1.10 mmol/L or total $<$ 8.0 mg/dl**; preterm = **$<$ 1 mmol/L or total $<$ 7.0 mg/dl**.

Phosphorus – normal **5-7.8 mg/dl**; ~85% in skeleton; low at birth then rises; higher in formula-fed babies. **Inverse relationship between serum calcium and phosphorus.**

Magnesium – normal **1.6-2.8 mg/dl**. Hypomagnesaemia (< 1.6, symptomatic < 1.2) -> irritability, tremors, seizures; **frequently accompanies hypocalcaemia – you must correct the magnesium to correct the calcium.** Hypermagnesaemia (> 2.8) usually from excess administration; treat with hydration $\pm$ diuretics.

Thyroid profile

- **TSH is part of the newborn screen.**
- Maternal **hypothyroidism** alone is not an indication; past maternal **hyperthyroidism** -> add free T4 and TSH.
- Part of the **prolonged jaundice** work-up and in **failure to thrive**. Always check lab-specific ranges.

Bilirubin & liver function

- Serum bilirubin – monitored as part of jaundice management.
- **LFTs** indicated in all sick babies, in prolonged jaundice, if **direct bilirubin is raised**, or as UTI work-up; monitored in preterm babies on **TPN** (TPN-induced liver injury is a common cause of abnormal LFTs).

Components of liver function:

- **Synthetic** – serum **albumin** and **coagulation profile**
- **Liver cell injury** – **AST/ALT** and **LDH**
- **Cholestasis** – **Gamma-GT, alkaline phosphatase, direct bilirubin**

Imaging studies

- **Cranial ultrasound** – non-invasive; assesses **intraventricular haemorrhage, hypoxic-ischaemic damage, structural brain abnormalities.**
- **Chest X-ray** – RDS, meconium aspiration, pneumonia.

- **Echocardiography** – congenital heart defects, pulmonary hypertension, **PDA.**
- **Abdominal X-ray & ultrasound** – GI anomalies, **necrotising enterocolitis (NEC).**

References

Lecture material of Dr Ijeoma Stella; standard neonatology references.